1. Statement of the problem

Let H denote the hyperbolic plane lattice and let E_8 denote the negative definite E_8 root lattice.

Let $\pi\colon \mathcal{X} \to \Delta$ be a Tyurin degeneration of K3 surfaces, with central fibre $X_0 = V_1 \cup V_2$, which is H-polarised in the following sense (this corrects [GT24, Definition 3.20] as it appears in the arXiv version, which omits the final condition in (2)):

(1) (monodromy invariance) over the punctured disc Δ^* there is a trivial local subsystem $\mathcal{L}$ of $R^2\pi_*\mathbb{Z}$ with fibre H, so that for each $p \in \Delta^*$ the fibre $\mathcal{L}_p \subset H^2(X_p;\mathbb{Z})$ defines an H-quasipolarisation of X_p,

(2) (positivity) there exists an effective, flat, and nef Cartier divisor A on $\mathcal{X}$ such that the restriction A_p of A to any fibre X_p over $p \in \Delta^*$ is nef and big and satisfies $[A_p] \in \mathcal{L}_p$. Moreover, the restriction of A to V_i satisfies $(A|_{V_i})^2 > 0$ for each $i \in \{1,2\}$.

Assume that this degeneration is very general, in the sense that the image of $0 \in \Delta$ under the map from Δ to the compactified moduli space of H-polarized K3 surfaces defined by Alexeev-Brunyate-Engel [ABE22] is not contained in any Noether-Lefschetz locus.

Let Y be a very general $H \oplus E_8 \oplus E_8$-quasipolarised K3 surface. Then there exists an elliptic fibration $\pi\colon Y \to \mathbb{P}^1$ along with an allowable loop $\gamma \subset \mathbb{P}^1$ (in the sense of [GT24, Definition 4.11]), such that the $H \oplus E_8 \oplus E_8$-quasipolarisation is compatible with π and γ (in the sense of [GT24, Definition 4.21]) and such that $\mathcal{X} \to \Delta$ and $\pi\colon Y \to \mathbb{P}^1$ are a mirror pair in the sense of [GT24, Definition 5.1].

References

[ABE22] V. Alexeev, A. Brunyate, and P. Engel, *Compactifications of moduli of elliptic K3 surfaces: Stable pair and toroidal*, Geom. Topol. **26** (2022), no. 8, 3525–3588.

[GT24] L. Giovenzana and A. Thompson, *Degenerations and fibrations of K3 surfaces: Lattice polarisations and mirror symmetry*, Preprint, May 2024, `arXiv:2405.12009`.